CHAPTER I

THE PROBLEM AND ITS SCOPE

Introduction

Digital technology and its continuous innovation have reshaped the academic information and research resources management process in the context of a university. Most universities are implementing digital management systems to streamline academic and administrative processes. As stated by Phuong and Hai (2026), digital transformation has boosted the performance of higher education institutions by integrating digital technology into their operations, making institutional work procedures more effective. Within this transition, institutional repositories contribute remarkably to effectively collecting and preserving academic works and providing access to the general community. Further, incorporating advanced technologies such as semantic search within academic repositories supports users accessing information by understanding the context and actual intent of queries, rather than finding results based on keyword matches, leading to the search being more precise and fast in academic repositories (Lubis et al., 2026).

Capstone projects in higher educational institutions constitute a significant academic document representing the cumulative learning, capstone competence, and creative thinking skills of the student. As the volume of produced documents grows year by year, the problems related to storing, accessing and searching those works are escalating within institutional repositories. However, the keyword search method relies on exact matches, which limits its efficiency in finding related studies; therefore, searching titles and comparing documents is done manually. Furthermore, semantic search-based methods have seen significant advancement with proven effectiveness in retrieving conceptually associated papers and have proved their importance in digital repositories for higher education (Musa et al. 2026).

Although digital repositories have provided a useful tool, many universities still face challenges in measuring the similarity of capstone titles and research objectives, as well as identifying research trends from previously completed studies. As the number of capstone reports continues to grow, manually reviewing and comparing research proposals becomes increasingly time-consuming and inefficient. Without an automated similarity detection tool, evaluating the relatedness of proposed studies becomes more